

Quality-Adaptive Multi-Task Oculomics: Extending Reti-Pioneer with Learnable Fusion, Calibration, and Public-Cohort Validation

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Target
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> One-sentence argument. Reti-Pioneer showed that frozen foundation features plus quality-aware fusion can screen systemic disease from color fundus photographs (CFPs); we keep that skeleton, replace fixed quality weights with learnable quality routing, replace independent binary heads with a shared multi-task head, and evaluate calibration / decision-curve utility and cross-dataset generalization on public cohorts reviewers can download – without claiming UK Biobank-comparable area under the receiver operating characteristic curve (AUROC) from synthetic demo caches.

Abstract

Endocrine and metabolic disease screening from retinal imaging (oculomics) remains limited by image-quality heterogeneity, single-task training, and poorly calibrated risk scores. Reti-Pioneer (Zhang et al., **Nature Medicine**, 2026) demonstrated that frozen vision foundation models with quality-aware bilinear fusion can screen six systemic conditions on UK Biobank (UKB) and hospital cohorts, but uses fixed quality routing, trains independent binary heads, and reports discrimination without treating calibration and decision-curve analysis (DCA) as primary utility. Here we propose a methods extension that (i) unfreezes quality routing (learnable_q), (ii) shares a multi-label head across correlated ocular/systemic labels, and (iii) reports expected calibration error (ECE), Brier score, and net benefit under temperature scaling. We specify patient-level evaluation on publicly downloadable Ocular Disease Intelligent Recognition (ODIR-5K), Brazilian Multilabel Ophthalmological Dataset (BRSET), and Retinal Fundus Multi-disease Image Dataset (RFMiD). Results tables with clinical AUROCs are marked 待补充 until real images and foundation features replace synthetic caches. Pipeline sanity metrics on synthetic features are reported only in the companion research report and are not manuscript claims.

Keywords: oculomics; fundus photography; quality-aware fusion; multi-task learning; calibration; domain generalization; Reti-Pioneer

1. Introduction

Color fundus photography is widely available in community eye care and can encode microvascular signatures of systemic disease. Foundation models such as RETFound (Zhou et al., **Nature**, 2023) provide transferable retinal representations, while quality assessment toolkits such as EyeQ support handling of imperfect acquisitions. Reti-Pioneer integrated frozen RETFound, Swin Transformer V2-B, and Vision Mamba-S with quality-aware fusion and metadata, reporting internal-test AUROCs of 0.699–0.833 across six diseases and demonstrating a primary-care silent trial.

Despite this progress, three methodological gaps remain actionable without UKB access. First, Reti-Pioneer freezes quality weights to good=1 / usable=0.5 / bad=0, which may be suboptimal when quality label distributions differ across cameras and sites. Second, training six

independent binary models ignores label correlation (e.g., diabetes-related and hypertensive findings) that multi-label ocular datasets naturally provide. Third, screening deployment needs calibrated probabilities and decision-curve net benefit, not AUROC alone.

Contributions.

1. A Reti-Pioneer-compatible learnable quality routing module initialized at the paper's fixed weights.

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Boundary. We do not claim UKB T2DM AUROC 0.833 replication; we do not treat synthetic feature-cache AUROCs as clinical evidence; gout / osteoporosis / hyperlipidemia / thyroid remain UKB-only until public gold standards exist.

2. Related work

Foundation models for CFP. RETFound and related retinal foundation models reduce labeled-data needs for ocular and oculomic tasks. Reti-Pioneer ensembles multiple frozen backbones rather than proposing a new foundation model; our work inherits that stance.

Quality-aware analysis. EyeQ-style classifiers produce good/usable/bad probabilities. Prior screening studies show community CFPs are often degraded; Reti-Pioneer's bilinear quality fusion is a strong baseline. Learnable routing is a bounded refinement, not a new quality taxonomy.

Multi-label fundus datasets. ODIR-5K, RFMiD, and BRSET provide multi-label ocular (and some systemic) annotations downloadable by reviewers. They are not UKB ICD endpoints; Methods must state label semantics explicitly.

Calibration and decision curves. Temperature scaling, EOE, and DCA are standard for clinical utility assessment. RETFound-enhanced community screening studies have used DCA; we adopt the same utility language for public oculomics baselines.

Positioning. This is a methods paper on top of Reti-Pioneer, not a competitor foundation model and not a Nature Medicine-scale clinical claim without biobank data.

3. Methods

3.1 Inputs

Paired or single CFPs, metadata (age, sex; weight/ethnicity padded when missing), and a 3-way quality probability vector. Patient IDs define folds so both eyes of one person stay in one split.

3.2 Frozen backbones and feature cache

Following Reti-Pioneer, we freeze RETFound / Swin V2-B / Vision Mamba-S and train on pre-extracted features (fast_mode). Public caches are built via prepare_public_npz.py then extract_features.py.

3.3 Quality routing

QualityAware implements bilinear fusion. With learnable_q=False, q_fc is frozen at (1, 0.5, 0). With learnable_q=True, the same initialization is used but q_fc is trainable. Optional supervised quality on BRSET focus/illumination/artifact labels is deferred (待补充).

3.4 Multi-task head

ComplexModel supports num_classes=K. Joint BCE (with optional pos_weight) shares the fused representation across labels. Independent single-disease training remains the Reti-Pioneer-clone control.

3.5 Ensemble

Train-time softmax soft voting and eval-time max across three backbone heads are retained; temperature-scaled mean is an alternative (--ensemble temp_mean).

3.6 Calibration and decision utility

Temperature T is fit on validation logits. We report AUROC / average precision (AP), ECE, Brier, sensitivity at high specificity, and net benefit at threshold 0.10 versus treat-all / treat-none.

3.7 Training protocol

Learning rate 1e-4 (config default), frozen backbones, patient-level split, seeds {42,43,44} for final tables (待补充 on real data). Ablation arms: baseline, learnable_q, multitask, full.

3.8 Label mapping (pre-registered)

Head · ODIR · BRSET · Honesty

4. Experiments (protocol)

- Datasets: ODIR-5K, BRSET, RFMiD (download status: 待补充 for full images).

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Honesty rule: never place synthetic/demo AUROC in submission tables.

5. Results

Clinical result tables: 待补充 (require real ODIR/BRSET/RFMiD pixels + foundation feature extraction).

For engineering verification only, a synthetic ablation matrix exists in results/ablation_summary.csv (see companion report). Those values are pipeline sanity, typically near chance on tiny synthetic caches, and must not be compared to Reti-Pioneer's 0.699-0.833 UKB internal AUROCs.

Expected qualitative story after real features (hypothesis, not claim): modest AUROC gains from multi-task correlation; larger ECE reductions from temperature scaling; quality routing helps

more when true quality labels exist (BRSET); cross-domain drop remains large and must be discussed.

6. Discussion

Public ocular+systemic labels enable reproducible methods claims that UKB-gated papers cannot always support. Transferring Reti-Pioneer’s skeleton to ODIR/BRSET forces explicit label semantics and domain-shift reporting. Fairness and calibration should be first-class, matching the authors’ own limitations discussion.

Figure reading contract (for companion report figures; manuscript clinical panels 待补充). Figure 1 is an information-flow schematic only (yellow = learnable quality fusion; green = multi-task + calibration/DCA). Figures 2-4 currently plot SYNTHETIC feature-cache metrics near chance with tiny *n*; they justify engineering readiness and the *protocol* of reporting ECE and cross-dataset drop, not clinical superiority over Reti-Pioneer’s UKB AUROCs.

Honest novelty phrasing. Prefer: “we unfreeze quality routing initialized at Reti-Pioneer’s fixed weights and evaluate jointly with multi-label heads and calibration/DCA on public cohorts.” Avoid: “we outperform Reti-Pioneer” or “we achieve AUROC X on T2DM” without the original UKB endpoint and cohort.

7. Limitations

No UKB / SEED access in this workspace; label mismatch vs ICD systemic endpoints; no prospective trial; Vision Mamba optional on Windows; EyeQ weights external; synthetic caches for CI only.

8. Reproducibility

Code, configs (configs/ablation_*.yaml), scripts/run_ablations.py, patient-level split, and docs/REPRODUCIBILITY.md. BRSET requires PhysioNet credentialing.

powershell

pytho

References (seed list; expand on real submission)

1. Zhang et al. AI framework for multidisease detection via retinal imaging. *Nat Med* (2026). doi:10.1038/s41591-026-04359-w

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Assumptions or missing inputs

- Real ODIR/BRSET/RFMiD downloads and GPU feature extraction not completed in this workspace.

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